Chapter 7 Systems control engineering Basic Labs

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Chapter 1 Report Assignment First Week : State equation and feedback control

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In Chapter 1 in the first week, we learn the mechanism of a spring-mass-damper system, feedback control on spring-mass-damper system and express state equations for linear systems. The feedback control is done by changing the values of feedback gain k_p (proportional constant) and k_d (damping constant) to reach a certain eigenvalue of the system. k_p and k_d are free constants because we can design the controller. We can do the other way as well through pole placement: set the system eigenvalue first, express the eigenvalue as a pole in the complex plane (real and imaginary part), and then from that determine the value of the feedback gain settings k_p and k_d

Lab equipment:

1. A mass placed on a horizontal plane such that it moves smoothly in the horizontal direction on a linear bearing, and a viscosity damper equal to an applied spring force in the distance horizontal to that mass. The mass section acts as a DC linear motor (voice 42 coil motor) needle, and can freely apply a thrust that is proportional to the current flowing through the primary side voice coil motor.
2. Oscilloscope
3. MATLAB simulations

Reporting assignment 1.1

1. The natural angular frequency $\omega_0 = \sqrt{k/m} = \sqrt{\frac{1.2X10^3}{1.1}}$ Hz = 33.02 Hz

   Natural frequency $\omega = \frac{\omega_0}{2\pi}$ = 5.256 Hz = 5.26 Hz

2. If the damping ratio γ is larger or equal to 1, there will be no oscillation.

   $\gamma = \frac{c}{2m\omega_0}$

If $\gamma \geq 1$, then $c \geq 2m\omega$

c ≥ 2 (1.1) (5.26)

c ≥ 11.572 Nm/s

3. The solution x = $e^{\frac{-ct}{2m}}$C cos ($\frac{\sqrt{4mk-c^2}}{2m}t + \phi$)

If we take the limit as time passes into infinity for the first multiplier $e^{\frac{-ct}{2m}}$,

$\lim_{t\to\infty} e^{\frac{-ct}{2m}} = 0$

So, the displacement of a damper always asymptotically approaches 0 after sufficient time has elapsed from the disposition of the damper and the string.

Reporting assignment 1.2

1. For m =1, the matrix A of the state equation= $\begin{bmatrix} 0 & 1 \\ -k & -c \end{bmatrix}$

   The characteristic equation which contains the eigenvalues can be found from det (A-λI)=0

   Characteristic equation $\lambda^2 + c\lambda + k = 0$

a) When the characteristic root becomes $\lambda_1 = 2i \; and \; \lambda_2 = -2i$, the characteristic equation becomes
-4+c(2i)+k=0 and
-4-c(2i)+k=0
Solving these simultaneous equations: c=0 and k=4
In terms of physics from the perspective of the spring force and the damping force, there is no damping force but there is a restorative spring force which is in the direction opposite to the displacement.

b) If $\lambda_1 = -1 \; and \; \lambda_2 = -1$, the characteristic equation becomes 1+c+k=0 and 1-c-k=0
Solving these simultaneous equations: c=1 and k=-2
In terms of physics, there is damping force opposite to the displacement and a spring force that is in the same direction as the displacement. The damping force tries

c) If λ_1 = 1 + 2i $and \; \lambda_2 = 1 - 2i$, the characteristic equation becomes
(-3+4i)+(1+2i)+k=0 and (-3+4i)-(1-2i)+k=0
Solving these simultaneous equations: c=-2 and k=2
In terms of physics, there is a damping force in the same direction as the displacement and a spring force that is opposite direction to the displacement.

d) If $\lambda_1 = -1 \; and \; \lambda_2 = -2$, the characterisitic equation becomes
1-c+k=0 and 4-2c+k=0
c=3 and k=2
In terms of physics, the damping force and spring force are both opposite direction to the displacement.

Sign convention of the spring force and the damping force:
When the spring is stretched or extended from its free length it will exert a negative force between the connected bodies (between the spring and the mass) . Likewise when the spring is compressed it will exert a positive force.

When the starting force of a spring is positive it will push apart the two connected bodies. Likewise if the starting force of a spring is negative it will pull together the two connected bodies.

The sign convention of the damping force in the spring is that the damping force always opposes the separation velocity between the two end points of the spring. If the spring is increasing in length then the damping force will push to try to decrease the spring length. Likewise if the spring is decreasing

in length then the damping force will push to try to increase the spring length.

Reporting assignment 1.3

1. A= $\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$

   $\frac{dx}{dt} = A\boldsymbol{x}$ with **x**= $\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$

   $= \begin{bmatrix} \frac{dx}{dt} \\ -\frac{k}{m}x - \frac{c}{m\frac{dx}{dt}} \end{bmatrix}$

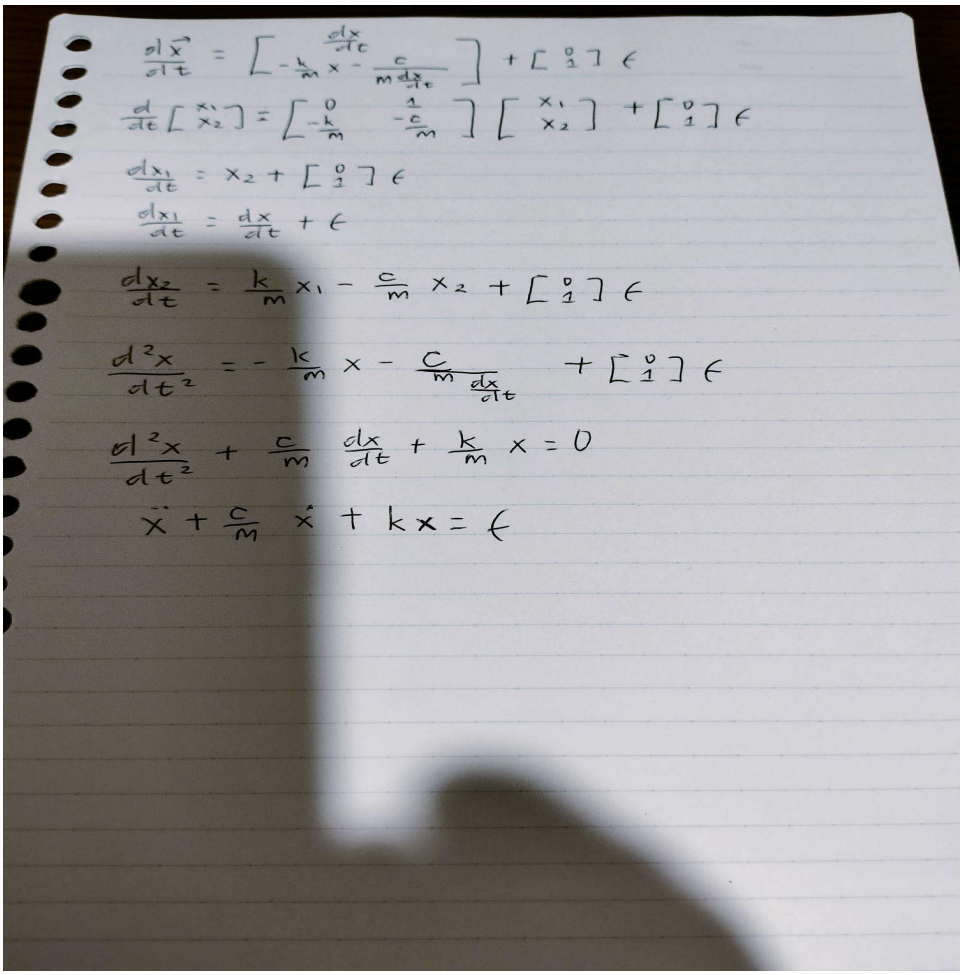
   $$\ddot{x} + (\frac{c}{m})\dot{x} + (\frac{k}{m}) = 0$$
   $$\mathrm{m}\ddot{x} + c\dot{x} + kx = 0$$

   If we use solution x= $e^{\lambda t}$ and substitute it to the former equation, we'll get the secular equation (characteristic equation)
   $m\lambda^2$+cλ+k =0

2. $\frac{dx}{dt} = A\boldsymbol{x} + \boldsymbol{b}u$ with A= $\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$, **x**= $\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$, **b**= $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and u =f (Equation (1.14)

   $$\frac{dx}{dt} = \begin{bmatrix} \frac{dx}{dt} \\ -\frac{k}{m}x - \frac{c}{m\frac{dx}{dt}} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f$$

   By expanding and solving this equation, this matches equation (1.3) $\mathrm{m}\ddot{x} + c\dot{x} + kx = f$ (See the figure below)



$$\frac{d\vec{x}}{dt} = \begin{bmatrix} \frac{dx}{dt} \\ -\frac{k}{m}x - \frac{c}{m}\frac{dx}{dt} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\epsilon$$

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\epsilon$$

$$\frac{dx_1}{dt} = x_2 + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\epsilon$$

$$\frac{dx_1}{dt} = \frac{dx}{dt} + \epsilon$$

$$\frac{dx_2}{dt} = \frac{k}{m}x_1 - \frac{c}{m}x_2 + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\epsilon$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x - \frac{c}{m}\frac{dx}{dt} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}\epsilon$$

$$\frac{d^2x}{dt^2} + \frac{c}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$$

$$\ddot{x} + \frac{c}{m}\dot{x} + kx = \epsilon$$

Hence, the secular equation matches equation (1.14)

Reporting assignment 1.4

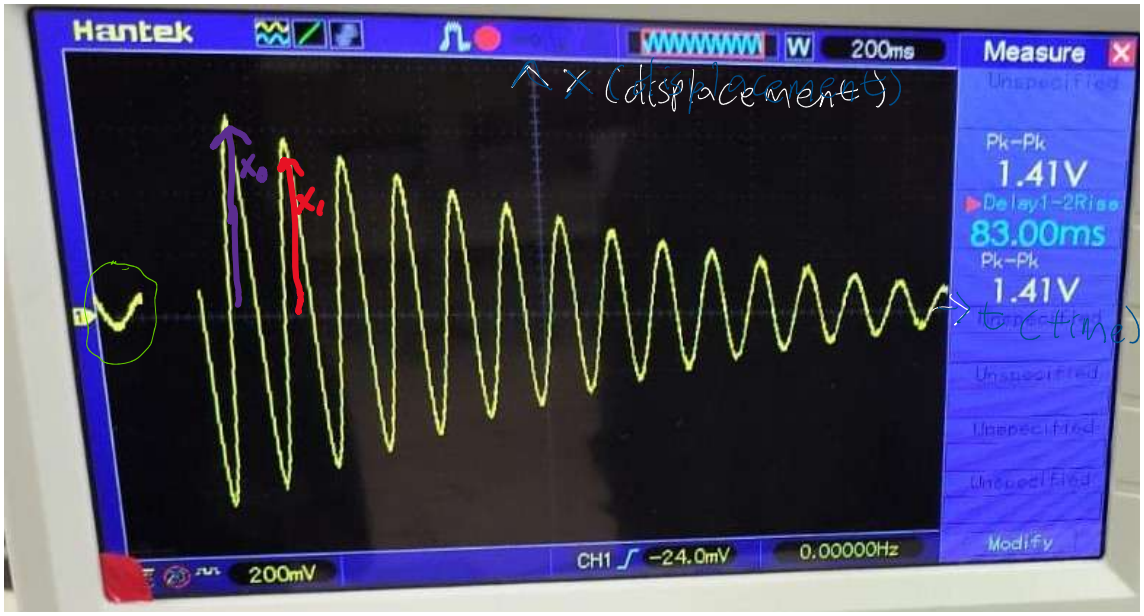


Figure 1 Oscilloscope time-history waveform

The first wave with broken end (marked with green circle) is not included in this wave calculation for ease of calculation

So I will start from this point marked with purple, x_0

1. The system eigenvalues can be seen from the amplitude envelope (which will be the real part of the eigenvalues) and the frequency (which will be the imaginary part of the eigenvalues)

   Here, x= p e^{at} $(cosbt + \varphi)$ and $\frac{x_2}{x_1} = e^{a(t_2 - t_1)}$

   The system eigenvalues are $\lambda = a \pm ib$

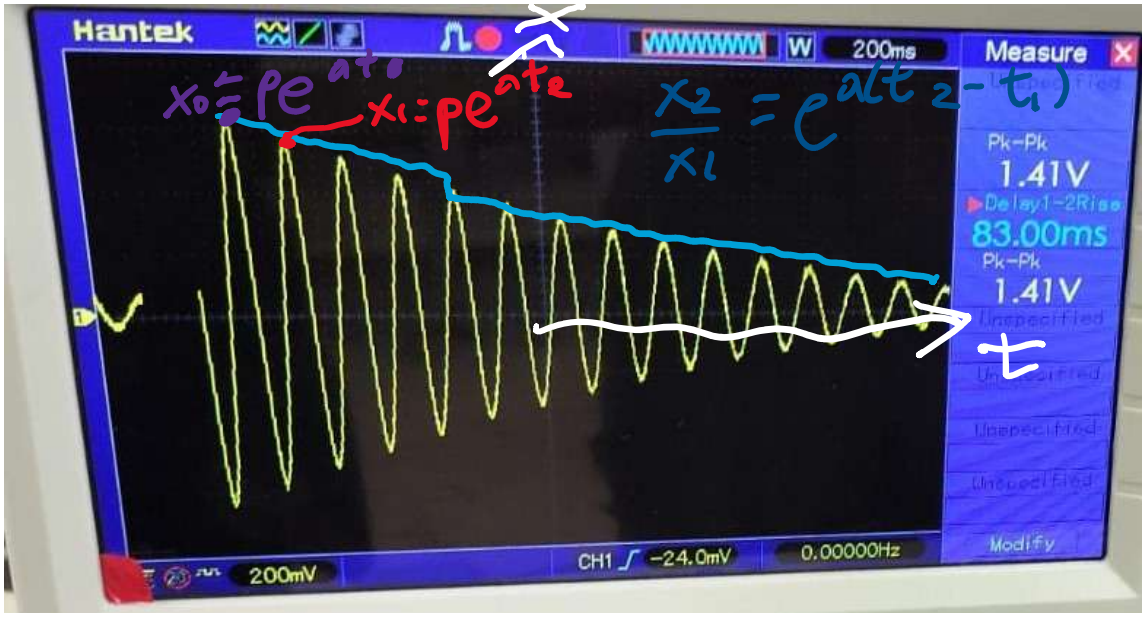


Figure 2

Voltage = 200 mV

From Figure 1, period T = 200 ms

T = $\frac{2\pi}{\omega_0}$, so $\omega_0 = 31.4\ Hz$

2. k= ${\omega_0}^2$ m = (31.4^2) (1.1) = 1085.66 N
3. There is a relationship between logarithmic decrement γ and δ

$$\gamma = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}}$$

where $\delta = \ln \frac{x_0}{x_1}$ and x_0, x_1 are amplitudes of two consecutive positive or negative peaks.

From Figure 1, $x_0 = (3.2)\ (10)\ (2\text{X}10^{-3})$ = 0.064 mm and $x_1 = (2.8)(10)(2 \times 10^{-3})$

= 0.056mm

$\gamma = \frac{0.134}{\sqrt{(0.134)^2 + (2\pi)^2}}$ = 0.021

c = 2m $\omega_0\ \gamma = 2(1.1)(31.4)\ (0.021)$

= 1.45068 =1.45 Ns/m

4. $\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} = \frac{-1.45 \pm \sqrt{(1.45)^2 - 4(1.1)(11085)}}{2(1.1)}$

= (-1.45 ± 69.1 i) / 2.2

$\lambda_{1,2} = 0.659 \pm 31.41i$

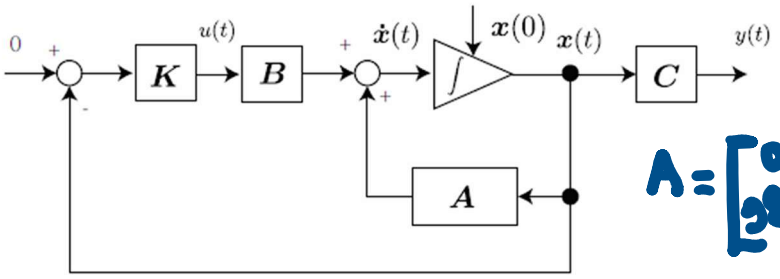
Reporting assignment 1.5

The state equation of the system is

$$\frac{d}{dt}\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} - K_p & \frac{-c}{m} - K_d \end{bmatrix} \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$$

So the state equation is $\frac{d}{dt}\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 986 - K_p & -1.31 - K_d \end{bmatrix}$

2） The block diagram is



$k = [K_p \; K_d]$

$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$A = \begin{bmatrix} 0 & 1 \\ 986 & -1.31 \end{bmatrix}$

Week 2 Report Chapter 2: Frequency response and PID control

In this chapter, we learned the frequency response when the periodic input expressed by a trigonometric function is applied to a linear system, graph the input-output relationship of frequency response as a bode diagram, and alter the gain constants and observe the frequency response and system characteristics in proportional control, integral control, and PID control

Lab Equipment:

1. Spring-mass-damper system as in Chapter 1/ Week 1
2. LCR circuit
3. Oscilloscope
4. Simulation in MATLAB

Reporting assignment 2.1

The figure of the bode diagram is shown in the graph paper.

Reporting assignment 2.2

The figure of the bode diagram of frequency response curve according to spring-mass-damper system, with the frequency increased in a 0.2 interval, and the gain and the phase difference measured, is shown on the graph paper. The mass section acts as a DC linear motor needle, and can freely apply thrust proportional to current flowing through the primary-side voice coil motor. The input voltage was 0.1 V. Respective relationships with the frequency are shown in the bode diagram.

For the gain and frequency response relationship graph, there is an outlier maximum spike at gain of 0.0023 V.

Reporting assignment 2.3

The figure of the bode diagram according to LCR circuit including an AC voltage source is shown in the graph paper, with the frequency increased in a 0.2 interval measured and a constant input voltage of 0.1 V, and the gain and the phase difference measured. The respective relationships with the frequency are shown in the bode diagram.

Reporting assignment 2.4

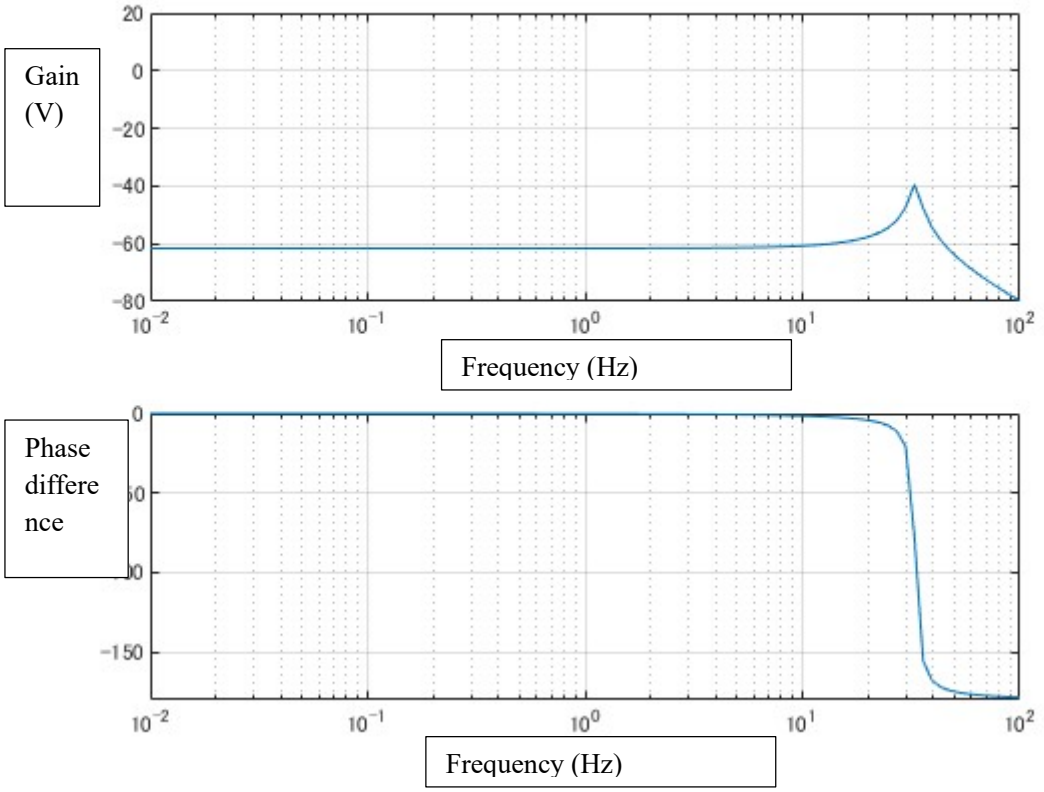


Figure 1: Bode diagram made in Practice Assignment 2.5

The upper graph of the bode diagram in Practice Assignment 2.5 (Figure 1), which shows the relationship between the gain and the frequency, looks similar to the bode diagrams in the bode diagrams of the lab experiments discussed in reporting assignments 2.2 and 2.3. The gain -frequency diagram made in Practice Assignment 2.5 has a steep peak, but the peak is steeper in Reporting Assignment 2.2.

However, the lower graph of the bode diagram, which shows the relationship between the frequency and the phase difference, shows a difference. All the diagrams show a steeping curve, however. The phase difference and frequency obtained from the creation of the spring-mass-damper system in reporting assignment 2.2 looks unusually different to the other bode diagrams. The phase difference- frequency response bode diagram in reporting assignment 2.3 shows an almost linear descent (negative slope)

Reporting assignment 2.5

Relationship between poles and responses based on the results obtained in Practice Assignment 2.9:

a) For (Kp, Ki, and Kd) = (500, 200, 500):
   All poles have negative real parts. Two poles have imaginary parts.
   From the graph, we can see the graph comparatively quickly reaching the target.
b) For (Kp, Ki, and Kd) = (500, 0, 500):
   There is a zero pole, and there is a negative real pole with no imaginary part:. From the graph by setting Ki to zero, the graph at first rises but then reaches a constant value that is lower than the target value and never reaches the target value
c) For (500, 200, 0):
   All poles have negative real part and no imaginary real part. By setting Kd to 0, the graph rapidly oscillates before reaching the target value
d) For (0,200,500):
   Pole 1: Pure imaginary number and Pole 2: negative real part with no imaginary part
   From the graph, by setting Kp to 0, the graph overshoots the target value
e) For (50, 20, 50):

   Pole one is zero, while two poles have negative imaginary and real part. The graph constantly rises but never reaches the target value.

   For the frequency response graph, it can be seen the importance of having both all three Kp, Ki, and Kd and all of them have to be large enough. Without proportion control gain Kp, the graph will overshoot the target value, without integral control gain Ki, the graph will never reach the target value, and without derivative control gain Kd, the graph will rapidly oscillate before reaching the target value. If all Kp, Ki, and Kd exist but are of small value, the graph will not oscillate but never reach the target value.

   If Ki is zero, the pole will have no imaginary part.

2. Comparing the results of integral control obtained from Practice 2.7 to PID control from Practice 2.9 :

In integral control, we notice that if we set Ki into an extreme value such as 25000, the pole will be in the right side of the complex plane. A system is stable if the root locus diagram poles are on the left side of the complex plane (real negative real parts). Whereas in PID control, the poles are either zero or on the left side of the complex plane. A system is stable if the root locus diagram poles are on the left side of the complex plane (real negative real parts).

Another reason , PID control is superior compared to only integral control because the integral control is much slower due to the former needing more parameters. (Integral control needing more parameters than proportional control)

3. Respective roles of the P, I, and D in PID control:

From the experiment:

For the frequency response graph obtained in Practice 2.9, it can be seen the importance of having both all three Kp, Ki, and Kd and all of them have to be large enough. Without proportion control gain Kp, the graph will overshoot the target value, without integral control gain Ki, the graph will never reach the target value, and without derivative control gain Kd, the graph will rapidly oscillate before reaching the target value. If all Kp, Ki, and Kd exist but are of small value, the graph will not oscillate but never reach the target value.

From theory:

a) P control : Proportional control

The reason why on-off control often gives rise to oscillations is that the system overreacts since a small change in the error makes the actuated variable change over the full range. This effect is avoided in proportional control, where the characteristic of the controller is proportional to the control error for small errors.

The signal produced by a proportional (P) controller is proportional to the control error. The steady-state performance and speed of response of a system with a proportional controller improve with increasing gain k_p. Increasing the controller gain may, however, decrease stability margins.

b) I control: Integral control

A controller with integral action has zero steady-state error. The gain of the integral control is K_i . Adding integral control, although improving the steady-state performance, may lead to oscillatory response (that is, reduced potential stability margin), which is usually undesirable. The catch is that there may not always be a steady state because the system may be oscillating.

c) D control: Derivative control

The stability of a system can be improved by the addition of a derivative action to the control algorithm. The derivative action provides an anticipatory effect that results in a damping of the system response. When the stability margin is increased in this way, it becomes possible to use a greater loop gain, thus improving speed of response and reducing steady-state error.

At time t, the proportional term depends on the instantaneous value of the error. The integral portion of the feedback is based on the integral of the error up to time t (shaded portion). The derivative term provides an estimate of the growth or decay of the error over time by looking at the rate of change of the error.

The control action is thus a sum of three terms: the past as represented by the integral of the error, the present as represented by the proportional term and the future as represented by a linear extrapolation of the error (the derivative term). This form of feedback is called a proportional-integral-derivative (PID) controller.

Conclusion

- **Effect of feedback control**

In this lab, we learned and experimented with system feedback control of linear systems through lab machines and Matlab simulations. In the first week, we learned about spring-mass-damper system and linear system feedback control, altering free response through setting the values of Kp and Kd. In the second week, we learned about frequency response, expressing the input-output relationship of the frequency response and gain and phase difference through Bode Diagram, and experimented with the effect of Proportional control, Integral control, and PID control.

- **Role of each gain of PID controller**

Increasing the proportional gain (Kp) has the effect of proportionally increasing the control signal for the same level of error. The fact that the controller will "push" harder for a given level of error tends to cause the closed-loop system to react more quickly, but also to overshoot more. Another effect of increasing Kp is that it tends to reduce, but not eliminate, the steady-state error.

The addition of a derivative term to the controller (Kd) adds the ability of the controller to "anticipate" error. With simple proportional control, if Kp is fixed, the only way that the control will increase is if the error increases. With derivative control, the control signal can become large if the error begins sloping upward, even while the magnitude of the error is still relatively small. This anticipation tends to add damping to the system, thereby decreasing overshoot. The addition of a derivative term, however, has no effect on the steady-state error.

The addition of an integral term to the controller (Ki) tends to help reduce steady-state error. If there is a persistent, steady error, the integrator builds and builds, thereby increasing the control signal and driving the error down. A drawback of the integral

term, however, is that it can make the system more sluggish (and oscillatory) since when the error signal changes sign, it may take a while for the integrator to "unwind”.

Source and reference:

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